

SÉBASTIEN MOSSER

DR. SÉBASTIEN MOSSER
COMPUTING AND SOFTWARE DEPT.
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2010	Ph.D., Computer Science & Software Engineering, <i>Université de Nice</i> , France
2007	Engineering Degree, <i>École Polytechnique Universitaire de l'Université de Nice</i> , France
2004	<i>Diplôme d'Études Universitaires Générales</i> , <i>Université de Nice – Sophia Antipolis</i> , France

2026	<i>(in progress)</i> Engineering Leadership Excellence Academy, <i>McMaster University</i>
2023	Inclusive Excellence Leadership, Equity and Inclusion Office, <i>McMaster University</i>
2022	Instructional Skills Workshop, McPherson Institute, <i>McMaster University</i>
2022	Ethical Conduct for Research Involving Humans – TCPS 2, <i>CIHR/SSHRC/NSERC, Canada</i>
2022	McMaster University Ethics Tutorial, <i>McMaster Ethic Board Committee, Canada</i>
2021	Bias in Peer Review, <i>CIHR/SSHRC/NSERC, Canada</i>

2011 Postdoc, Software engineering, *Inria Lille-Nord Europe*, France

01/2022 – ...	Associate Professor with tenure , Faculty of Engineering, CAS department	
	Associate Chair , Dept. of Computing and Software (CAS)	<i>since 07/2024</i>
	Associate Director , McMaster Centre for Software Certification (McSCert)	<i>since 12/2024</i>
	Faculty member, McMaster Institute for Research on Aging (MIRA)	<i>since 01/2023</i>

06/25 - ...	Member, <i>Institute of Electrical and Electronics Engineers</i> (IEEE).
06/23 – ...	P.Eng., <i>Professional Engineers Ontario</i> (PEO)
12/21 – 05/23	P.Eng., <i>Ordre des Ingénieurs du Québec</i> (OIQ). (<i>License transferred, 05/23</i>).
01/21 – ...	Member, <i>Société Informatique de France</i> (Academic Society).
09/18 – ...	Member, <i>Association for Computing Machinery</i> (ACM)

EMPLOYMENT HISTORY

Academic

01/22 – ...	Associate Professor with tenure, Faculty of Engineering, CAS department, McMaster, Canada
06/21 – 12/22	Associate Professor with tenure, Faculty of Science, <i>Université du Québec à Montréal</i> , Canada (Administrative leave of absence from 01/22 to 12/22)
01/19 – 05/21	Associate Professor, tenure-track, Faculty of Science, <i>Université du Québec à Montréal</i> , Canada
09/13 – 12/18	<i>Maître de conférences</i> , tenured, Université Côte d'Azur, Nice, France
09/12 – 08/13	<i>Maître de conférences stagiaire</i> , tenure track, Université Côte d'Azur, Nice, France
09/11 – 08/12	Research Scientist, SINTEF, Oslo, Norway
11/10 – 09/11	Postdoctoral fellow, Inria Lille – Nord Europe, Villeneuve d'Ascq, France
09/07 – 10/10	Ph.D. fellow, Université de Nice, Nice, France
09/07 – 10/10	<i>Moniteur de l'enseignement supérieur</i> , Université de Nice, Nice, France

SCHOLARLY AND PROFESSIONAL ACTIVITIES

Editorial Boards

06/25 – ...	Co-Editor in Chief. <i>Journal of Object Technology</i>	(AITO).
05/25 – 12/26	Co-Editor. <i>Software Certification: Challenges and Promising Directions</i> .	(Springer LNCS)
01/22 – 05/25	Editorial Board Member. <i>Journal of Object Technology</i>	(AITO).

Grant & Personnel Committees

10/25 – ...	Joint Prize Selection Committee, NSERC	Committee member
06/22 – ...	GDR GPL (CNRS). French Best PhD Award in Comp. Sci./Soft. Eng.	Committee member.
10/22 – ...	CS-CAN Excellence in Teaching award.	Committee member.
06/22 – 05/24	NSERC 1507 Discovery Grant Committee.	Co-chair (Soft. Eng.)
06/21 – 05/22	NSERC 1507 Discovery Grant Committee.	Committee member
01/20 – 12/21	<i>Fond de Recherche Québécois (FRQNT)</i> . PhD grant awards.	Committee member.
01/19 – 12/20	NSERC USRA. Internal committee at UQAM.	Committee member.

Executive Positions

01/26 - ...	Strategic Committee, CNRS GdR SciLog <i>GdR SciLog is a national instrument funded by the Centre National de la Recherche Scientifique (CNRS) in France to design and implement a strategic orientation and national research program for research in Software Engineering and Computer Science at the national level. It shapes the country's research policy and calls from national funding agencies.</i>	Executive Board
11/25 - ...	Software Engineering Awareness Network, CNRS. <i>The Software Engineering Awareness Network is an international network funded for 5 years (2025-2030) by CNRS Informatics under the International Research Network program</i>	Governance Board
01/22 - ...	McMaster Centre for Software Certification (McSCert). <i>McSCert is a leading research centre in Software Safety, regrouping five professors, 3 research engineers, and around 40 highly qualified personnel (HQP) yearly.</i>	Executive Board
01/21 – 12/24	Working group on Model-Driven Engineering & Education, MDENet.	Founding member.
01/16 – 12/18	Steering committee for hiring (CPRH), Université Côte d'Azur.	Nominated member
01/08 – 12/10	Steering committee, Action IDM (CNRS, France).	Student member
11/07 – 12/15	Steering Committee. <i>La Nuit de l'Info</i> (Univ. competition, ~50k€/y).	Founding member.

Journal Referee

Journal of Software and System Modelling (SoSyM, Springer).	28 reviews
Journal of Object Technology (JOT, AITO).	6 reviews
Empirical Software Engineering (ESE, Springer).	6 reviews
Transactions on Cloud Computing (TCC, IEEE).	2 reviews
Journal of Computer Languages (COLA, Elsevier).	2 reviews
Journal of Systems and Software (JSS, Elsevier).	3 reviews

Software Quality Journal (SQJ, Springer).	2 reviews
Journal of the Internet of Things (IoT, IEEE).	1 review
Automated Software Engineering (ASc, Springer)	1 review

External Grant Reviews

2024	Linz Institute of Technology, (Ministry of Education, State of Upper Austria, JKU). Austria.
2023	Agence Nationale de la Recherche (ANR, France). Early Career Research Program (JCJC)
2020	Agence Nationale de la Recherche (ANR, France). Early Career Research Program (JCJC)
2020	<i>Institut Mines-Telecom Atlantique</i> (IMT, France). Research Centre creation committee.
2017	NSERC 1507 Discovery Grants. External Reviewer.
2011	Agence Nationale de la Recherche (ANR, France). Industrial Transfer Project.

Internal Grant Reviews

2024 - ...	McMaster Institute for Research on Aging (MIRA). Labarge Postdoctoral Fellow.
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Conference Organization Committees

2026	2 nd International School on Foundations and Advances of Model-Based Engineering (FAME 2026), co-located with the 29 th ACM/IEEE International Conference on Model Driven Engineering Languages and Systems. <i>Co-chair</i> . Malaga, Spain.
2026	29 th IEEE International Symposium on Real-time Distributed Computing (ISORC 2026). <i>General Chair</i> . Hamilton, Ontario, Canada.
2025	28 th ACM/IEEE International Conference on Model Driven Engineering Languages and Systems (MODELS). <i>Program Committee co-chair (NIER)</i> . Grand Rapids, Michigan.
2025	21 st European Conference on Modelling Foundations and Applications (ECMFA 2025). <i>Program Committee co-chair</i> . Koblenz, Germany.
2025	International Joint Conferences On Theory and Practice of Software (ETAPS 2025). <i>Publicity co-chair</i> . Hamilton, Canada.
2024	27 th ACM/IEEE International Conference on Model Driven Engineering Languages and Systems (MODELS). <i>Doctoral Symposium co-chair</i> . Wien, Austria.
2024	18 th International Working Conference on Variability Modelling of Software-Intensive Systems (VaMoS). <i>Co-chair of the Artifact Evaluation track (creation of the track)</i> .
2023	26 th ACM/IEEE International Conference on Model Driven Engineering Languages and Systems (MODELS). <i>Social Media & Publicity Chair</i> . Västerås, Sweden
2022	<i>École des Jeunes Chercheuses et Jeune Chercheurs en Programmation</i> (EJCP, National summer school for Ph.D. students in software engineering & programming languages). <i>Co-chair</i> . Virtual.
2022	25 th ACM/IEEE International Conference on Model Driven Engineering Languages and Systems (MODELS). <i>Conference chair</i> . Montréal, Canada
2021	<i>École des Jeunes Chercheuses et Jeune Chercheurs en Programmation</i> (EJCP, National summer school for Ph.D. students in software engineering & programming languages). <i>Co-chair</i> . Virtual.
2020	23 rd ACM/IEEE International Conference on Model Driven Engineering Languages and Systems (MODELS). <i>Virtualization chair and Student Volunteers chair</i> . Virtual
2019	41 st ACM/IEEE International Conference on Software Engineering. <i>Accommodation chair</i> . Montréal, Canada.
2015	14 th ACM International Conference on MODULARITY. <i>Social Media chair</i> . Fort Collins, CO, USA.
2014	8 th International Workshop on Variability Modelling of Software-intensive Systems (VaMoS). <i>Organization committee</i> . Nice, France.
2012	3 rd IEEE World Congress on SERVICES. <i>Career development chair</i> . Honolulu, Hawaii, USA.
2011	3 ^{ème} journées nationales du GdR GPL (National conf.). <i>Organization committee</i> . Lille, France

Conference Program Committees

2025	19 th International Working Conference on Variability Modelling of Software-Intensive Systems (VaMoS). <i>PC member</i> .
2024	27 th ACM/IEEE International Conference on Model Driven Engineering Languages and Systems (MODELS). <i>Foundation track</i>

2023	26 th ACM/IEEE International Conference on Model Driven Engineering Languages and Systems (MODELS). <i>Foundation track & Doctoral Symposium mentor</i>
2022	25 th ACM/IEEE International Conference on Model Driven Engineering Languages and Systems (MODELS). <i>Foundation track</i> .
2021	43 rd ACM/IEEE International Conference on Software Engineering (ICSE). <i>Artifact evaluation track</i> .
2021	24 th ACM/IEEE International Conference on Model Driven Engineering Languages and Systems (MODELS). <i>Foundation track & Educator Symposium track</i> .
2020	24 th ACM International Systems and Software Product Line Conference (SPLC). <i>Research track</i> .
2020	23 rd ACM/IEEE International Conference on Model Driven Engineering Languages and Systems (MODELS). <i>Doctoral symposium track</i> .
2019	13 th IEEE International Conference on Research Challenges in Information Science (RCIS). <i>Doctoral Symposium track</i> .
2019	7 th International Conference on Model-Driven Engineering and Software Development (MODELSWARD). <i>Research track</i> .
2015 – ...	IEEE International Conference on Big Data (BigData). <i>Research track</i> .
2013 – 2021	IEEE International Conference on Web Services (ICWS). <i>Research track</i> .

Workshop Organization & Program Committees

2023	1st Workshop on Model-based Systems Engineering (co-located with MODELS). <i>Steering committee</i>
2022 – ...	Model-driven Requirement Engineering Workso (co-located with RE). <i>PC member</i> .
2023	3 rd International workshop on MDE for Smart IoT Systems (MESS, co-located with STAF). <i>Steering committee</i> .
2022 – ...	International Workshop on Requirements Engineering for Well-Being, Aging and Health (REWBAH). <i>PC member</i> .
2021 – ...	1 st International Workshop on Variability Management for Modern Technologies (co-located with SPLC). <i>Organization committee</i> .
2021 – 2022	International workshop on MDE for Smart IoT Systems (co-located with STAF). <i>Organization committee for the first two editions</i> .
2019 – 2022	International Workshop on DevOps modelling (co-located with Models). <i>Steering committee</i> .
2021	1 st International Workshop on Foundations and Practice of Visual Modeling (co-located with STAF). <i>PC member</i> .
2020	International Workshop on Software Engineering for the IoT (co-located with ICSE). <i>PC member</i> .
2018	International Workshop on Modeling for Microservices. <i>PC member</i> .
2016	2 nd International Workshop on Modularity in Modelling (co-located with <Programming>). <i>Organization committee</i> .
2014 – 2022	International workshop on Scalable Data Management (co-located with BigData). <i>PC member</i> .
2014	2nd International Workshop on Model-Driven Engineering on and for the Cloud. <i>PC Member</i> .
2013 – 2014	Nordic Workshop on Cloud Computing. <i>PC Member</i> .

AREAS OF INTEREST

Research

Software Engineering
Software Safety
Requirements Engineering
Domain-specific languages
Software design at scale

Teaching

Distributed Systems
Requirements Engineering
Software development & testing

Software design, requirements and modelling
Software construction and maintenance.

HONOURS

- 2025 **Visiting Professor Fellowship.** Université Toulouse Jean-Jaurès (UT2J), Toulouse, France.
- 2024 **Best Reviewer Award.** ACM/IEEE MODELS (27th edition)
- 2024 Nirmal Chaudhari (undergrad research assistant under my supervision) won the **silver medal** in the ACM Student Research Competition at MODELS for his research work done in my group.
- 2024 **IEEE TCSE Distinguished Synergy Award** awarded to McSCert during ICSE'24. (*co-recipient*)
This award is presented annually to a team for outstanding contributions that stand as a model in the software engineering community of effective partnership between industry and universities.
- 2022 Corinne Pulgar (MAcSc student under my supervision) won the **bronze medal** in the ACM Student Research Competition at MODELS for her research work done in my group (she was the only master student competing out of 14 participants).
- 2021 **Best Reviewer Award.** ACM/IEEE MODELS (24th edition)
- 2018 **Best Paper award,** 33rd ACM/SIGAPP Symposium on Applied Computing (SAC)
- 2015-2018 *Prime d'Encadrement Doctoral et de Recherche* (PEDR, National Council of Universities, France)
20k\$ salary bonus for excellence in research and doctoral supervision, evaluated nationally.

COURSES TAUGHT

Regular teaching assignment in the CAS department is two (2) undergraduate courses and one (1) graduate course per year. Associate Chairs get one undergrad teaching relief per year.

Undergraduate (at McMaster, since 2022)

Program

Year	Role/Title	Course Code/Title	Term	Section (C01, L01, T01)	% Taught	Enrolment	Duration	Additional Comments
2025	Instructor	SE/CS 3RA3	Fall	C01	100	188	1 term	–
2024	Instructor	SE/CS 3RA3	Fall	C01	100	204	1 term	–
2024	Instructor	SE 2AA4	Winter	C01	100	152	1 term	–
2023	Instructor	SE/CS 3RA3	Fall	C01	100	210	1 term	Creation
2023	Instructor	SE 2AA4	Winter	C01	100	162	1 term	Creation
2022	Instructor	SE 3XB3	Fall	C01	100	121	1 term	Creation

- SE 2AA4: *Software Design I – Introduction to Software Development*
- SE/CS 3RA3: *Software Requirements & Security Considerations*
- SE 3XB3: *Software Engineering Practice and Experience: Binding Theory to Practice*

Graduate (at McMaster, since Jan 2022)

Program

Year	Role/Title	Course Code/Title	Term	% Taught	Enrolment	Duration	Additional Comments
2025	Instructor	CAS 735*	Fall	100	11	1 term	Revamp.
2024	Instructor	CAS 735	Fall	100	21	1 term	–
2023	Instructor	CAS 735	Fall	100	19	1 term	–
2022	Instructor	CAS 735	Fall	100	20	1 term	Creation

- CAS 735: *(Micro-)Service Oriented Architectures*
(*). Starting Fall 2025, this course requires departmental permission to enroll, to ensure that students have the required background to take an advanced software engineering course. This explains the drop in enrollment.

CONTRIBUTIONS TO TEACHING PRACTICE

Pedagogic Innovation and/or Development of Technology-enhanced Learning

SFWRENG 2AA4 – Software Design I. In the second iteration of this course, I introduced two pedagogical innovations. First, a serious gaming approach where students can submit their code on a weekly basis and see how it compares in terms of delivered features to the others. Secondly, inspired by Dr. Sekerinsky approach in SE 3BB4, I introduced a “*Meaningful & Memorable*” feedback loop, where students can reflect on their learning objectives weekly and provide feedback to instructors and TAs, in both qualitative and quantitative ways.

UQAM/INF5153 – Software Design. To tackle the challenges of teaching software design in a COVID context, I re-designed the course into a “flipped classroom” one starting in Fall 2020. The “theoretical” part is published as openly accessible videos (creative commons license) on YouTube, representing more than 11 hours of content using the French language. The classroom time is used to work on case studies with small groups of students. Four universities now reuse this course material. Reusable material is available at the following URL: <https://conception-objet.github.io/> (FR)

Leadership in Delivery of Educational Programs

ISW During Summer 2022, I successfully completed the Instructional Skill Workshop offered by the McPherson Institute to reflect on course design and delivery,

Experiential learning. As an Associate Professor at *Université Côte d’Azur*, I oversaw the design and implementation of a “project-based”/experiential approach for software engineering courses in the department (involving redesigning ten courses in the program). Following up on this work, I was invited to give talks at several education conferences, and I consulted with 17 different universities in France on this topic.

Course/Curriculum Development

UQAM/INF600G – Designing tailored applications for the aging population. This course is designed as a collaboration between UQAM and a Human-Computer Interaction team in France (funded by Quebec’s research agency – FRQNT). We created the course to focus on designing and implementing software for the aging population. Students are confronted with multiple issues senior citizens face when using software and design adaptations to tackle these issues. Three institutions use the course (UQAM, Polytech Sophia, and IUT Nice – Côte d’Azur) and a collaboration with Toulouse (*Université Fédérale Toulouse – Midi Pyrénées*). URL: <https://ace-design.github.io/champlain/> (FR)

Development/Evaluation of Educational Materials and Programs

2014 – Engineering program evaluation. As part of a “school of engineering” in France, programs must be evaluated every six years by the *Commission du Titre d’Ingénieur* (CTI) at the national level (equivalent to CEAB). I led the software engineering part of the accreditation application (which was successfully renewed for six years).

Other

Guest Lectures. I am often invited to give “invited lectures” in the context of other courses. I regularly gave interventions dedicated to empirical software engineering (UCA, UQAM), software design (Toulouse), and microservice architectures (ETS) in other programs.

Continuous Training. As part of *Centre National de la Recherche Scientifique* (CNRS) initiative for engineers’ continuous training, I oversaw the courses “Agile software development” and “Business process modelling” for the *DevLog* national network in 2017.

SUPERVISORSHIPS

Note: Students in Montreal were initially only supervised by me but ended up co-supervised to ease my transition from UQAM to McMaster (2022), from an administrative point of view.

Master (thesis)

9 sole supervised, 3 co-supervised, 1 inactive.

Completed

Dates	Student's Name	Project Title	Department/Program	Supervisor	Co-Sup.
09/23 – 05/25	Stepan Bryantsev	The Single Source of Truth Paradigm as a Tool for Supporting Software Maintenance	CAS / MASc in Software Engineering	S. Mosser	
01/23 – 12/24	Alexandre Lachance	Static code analysis for P4	CAS / MASc in Software Engineering	S. Mosser	
09/22 – 09/24	Hunter (Hassan) Zaker	Migration to Microservice Architecture: A Semi-Automated Approach with Strangler Fig Methodology	CAS / MASc in Software Engineering	R. Paige	V. Pantelic S. Mosser
2017 – 2018	Günther Jungblunth	Developping scalable data-processing pipelines	MASc / UCA (France)	S. Mosser	
2015 – 2016	Benjamin Benni	A language-driven approach to software composition	MASc / UCA (France)	S. Mosser	
2013 – 2014	Cyril Cecchinell	Code generation applied to sensor networks	MASc / UCA (France)	S. Mosser	
2012 – 2013	Ivan Logre	User-centered dashboards for data collected by large scale sensor networks	MASc / UCA (France)	S. Mosser	
2011 – 2012	Eirik Brantzæg	CloudML, a DSL for model-based ealization of applications in the cloud	MSc / Universitetet i Oslo (Norway)	S. Mosser	

In progress

Dates	Student's Name	Project Title	Department/Program	Supervisor	Co-Sup.
09/25 – ...	Kalvin Khuu	Guidance system to support the reusability of FATES-related justifications in Machine Learning context.	CAS / MASc in Software Engineering	S. Mosser	
<i>Transfer from MEng program.</i>					
01/24 – ...	Arman Samiei	DevOps for software defined network	CAS / MASc in Software Engineering	S. Mosser	
<i>Coop internship, from 05/25 to 05/26</i>					

01/23 – ...	Azam Mahdipour (transfer from the M.Eng. program)	Single source of Truth for reverse engineering legacy software	CAS / MASc in Software Engineering	S. Mosser	V. Pantelic (McMaster)
		<i>Coop internship for one year, starting 09/24</i>			
09/21 – ...	Corinne Pulgar	Justification diagrams to evaluate the quality of DevOps pipelines	MASc / ETS Montréal	S. Mosser	F. Bordeleau (ETS Montréal)
		<i>Student is currently on a medical leave of absence (status: “all but dissertation”)</i>			

Inactive

Dates	Student's Name	Project Title	Department/Program	Supervisor	Co-Sup.
09/20 – 09/21	J.-P. Caissy	Reverse engineering of microservices architectures	Informatique UQAM. MSc in Computer Science	S. Mosser	
		<i>stopped on student's request because of COVID-19-related personal issues.</i>			

Doctoral

4 sole supervised, 7 co-supervised, 1 inactive.

Note: In the French system, “Maître de Conférences” (~Assistant/Associate Prof) cannot supervise doctoral students independently and needs a co-supervisor holding a Full Professorship, or pass a Habilitation thesis (HDR), usually after 10 years as “Maître de Conférences”. I had this status in France from 2012 to 2018. Based on my research activity and scientific results, the president of Université Côte d'Azur granted me two exceptional exemptions, allowing me to officially supervise the theses of B. Benni (2019) and I. Logre (2013) without co-supervision or HDR.

Completed

Dates	Student's Name	Project Title	Department/Program	Supervisor	Co-Sup.
2017 – 2020	Sébastien Bonnieux	Float for multidisciplinary monitoring of the marine environment. From business expertise to embedded codes	UCA (France). PhD in Earth and Universe Science	G. Nolet, M. Blay- Fornarino	S. Mosser
2016 – 2019	Sami Lazreg	Variability-intensive applications over highly configurable platforms: Early feasibility and optimality analysis	UCA (France). PhD in Computer Science	P. Collet	S. Mosser
2016 – 2019	Benjamin Benni	Enabling white-box reasonings on black- box composition operators in a domain- independent way	UCA (France). PhD in Computer Science	S. Mosser	
2014 – 2017	Cyril Cecchinell	DEPOSIT, an approach to model and deploy data collection policies on heterogeneous and shared sensor networks	UCA (France). PhD in Computer Science	S. Mosser	P. Collet
2013 – 2017	Ivan Logre	Preserving separation of concerns while integrating heterogeneous domains	UCA (France). PhD in Computer Science	S. Mosser	

		in software systems			
2010 – 2014	Alexandre Feugas	An agile, reliable, and minimalist approach to preserve the quality of service of business-processes based applications during their evolutions	University of Lille, PhD in Computer Science	L. Duchien	S. Mosser

In progress

Dates	Student's Name	Project Title	Department/Program	Supervisor	Co-Sup.
01/2025 - ...	Shyam Ravichandran	<i>Requirements engineering applied to Design thinking and HCI for older adults</i>	PhD in Soft. Eng McMaster University	S. Mosser	D. Geiskovitch
09/2024 - ...	Carlos Zegarra	<i>Capturing and Maintaining Design Decision over time in software pipelines</i>	PhD in Soft. Eng. McMaster University	S. Mosser	
09/2024 - ...	Nicolas Lacroix	<i>Supporting FATES properties identification in ML pipelines</i>	PhD in Comp. Science, Uni. Côte d'Azur (UCA)	M. Blay-Fornarino, F. Precioso	S. Mosser
09/2024 - ...	Nidhal Selmi	<i>Modelling trade-off decisions for critical systems.</i>	PhD in Comp. Science, Uni. Toulouse Jean Jaurès (UT2J)	J.-M. Bruel	S. Mosser

Inactive

Dates	Student's Name	Project Title	Department/Program	Supervisor	Co-Sup.
09/2021 - ...	Alexandra Lapointe-Boisvert	Functional testing to support software measurements	Informatique UQAM, PhD in Computer Science	S. Mosser	S. Trudel
		<i>Supervision paused as the candidate was promoted to an executive position in their company (Desjardins Bank), making the realization of a PhD thesis challenging.</i>			

Post-Doctoral/Fellowship

1 sole supervised, 0 co-supervised, 0 inactive.

In Progress

Dates	Student's Name	Project Title	Department/Program	Supervisor	Co-Sup.
09/2024 – 09/2026	Horacio Hoyos Rodriguez	Modelling variability for over the air update in the automotive industry	CAS / McMaster	S. Mosser	A. Wassying

Supervisory Committees

2024 - ... Lucas Dutton, PhD in Computer Science, McMaster University
 2024 Sheida Emadi, MASc in Software Engineering, McMaster University
 2023 - ... Mohamedreza Sabeghi, PhD in Software Engineering, McMaster University
 2023 - ... Lindsay White, PhD in Software Engineering, McMaster University.
 2022 Pete Michalsky. PhD in Software Engineering, McMaster University.
 2022 Saira Musa. PhD in Computer Science, McMaster University.
 2022 Naveen Ganesh Muralidharan. MASc in Software Engineering. McMaster University.

- 2019-2022 Hyacinth Ali (McGill University, Canada). Ph.D. in Software Engineering.
Modular combination and reuse of languages with perspective
- 2019-2023 Dimitri Prestat (UQAM, Canada). Ph.D. in Computer Science.
Formal detection of defaults in mobile applications.

Examination Committees

- 2025 Lucas Dutton, PhD in Computer Science, *Comprehensive Exam I.*
- 2025 Zhongyi Lu, PhD in Computer Science, University College Dublin. *Special external reviewer*
A Reputation-based Microservices Trust Model Using Similarity Domains (viva voce)
- 2024 Saira Musa, PhD in Computer Science. *Comprehensive Exam II.*
- 2024 Saira Musa, PhD in Computer Science. *Comprehensive Exam I (retake).*
- 2024 Saira Musa, PhD in Computer Science. *Comprehensive Exam I.*
- 2023 William Flageol. PhD in Computer Science, Concordia University. *External reviewer.*
- 2022 Devrim Tokcan. PhD in Software Engineering. *Comprehensive Exam.*
- 2022 Shams Alkhulaif. PhD in Software Engineering. *Comprehensive Exam.*
- 2021 Alexandre Rio (Université de Rennes, France). Ph.D in Computer Science. *External reviewer.*
Optimizing renewable energy usage: a digital twin for microgrids.
- 2021 Thibault Béziers La Fosse (Télécom Bretagne, France). Ph.D. in Computer Science.
External reviewer. Model-driven Method for Dynamic Analysis applied to Energy-Aware Software Engineering
- 2016 Thi-Mai-Anh Bui (Université Paris 6, France). Ph.D. in Computer Science. *External reviewer.*
Separation of concerns in epidemiology.

Master (non-thesis)

7 sole supervised, 0 co-supervised, 2 inactive.

Completed

Dates	Student's Name	Project Title	Department/Program	Supervisor	Co-Sup.
2024 – 2025	Waqar Assan	<i>Single source of Truth for maintenance of Microservice architecture</i>	M.Eng in Software Engineering / CAS	S. Mosser	
2023 – 2025	Jason Lyu	<i>Modelling High Availability good practices in Kubernetes using justification diagrams</i>	M.Eng in Software Engineering / CAS	S. Mosser	
<i>1-year coop internship during the completion of their MEng.</i>					
2022 – 2024	Kai Sun	<i>Justification diagram to evaluate quality of Jupyter notebooks</i>	M.Eng in Software Engineering / CAS	S. Mosser	
2022 – 2023	Deesha Patel	<i>Justification diagrams applied to CI pipelines</i>	M.Eng in Software Engineering / CAS	S. Mosser	
2021 – 2022	Mohamed Dramane Jean-Philippe Koïta	<i>Caractérisation des dépendances architecturales dans les architectures microservices</i>	UQAM / M.Eng.	S. Mosser	
2021 – 2022	Amine Soufyani	<i>Evolution et impact des technologies de déploiement dans les architectures orientées microservices</i>	UQAM / M.Eng.	S. Mosser	

In progress

Dates	Student's Name	Project Title	Department/Program	Supervisor	Co-Sup.
01/26 – ...	Sihui He	TBD	M.Eng in Software Engineering / CAS	S. Mosser	

Inactive

Dates	Student's Name	Project Title	Department/Program	Supervisor	Co-Sup.
09/24 – 08/25	Kalvin Khuu	TBD	M.Eng in Software Engineering / CAS	S. Mosser	
		Student transferred to MASc. program under my supervision.			
01/25 – 05/25	Houston Zhang	TBD	M.Eng in Software Engineering / CAS	S. Mosser	
		Student transferred to another supervisor to better align with their professional carrer.			

Research Interns supervision

- 2025** Baptiste Lacroix, MASc internship, Université Côte d'Azur. *Justification models for FATES properties in machine learning pipelines.*
 Darshan Shah, BEng internship, McMaster (Excellence in Research Award). *Justifying benchmarks for machine learning pipelines.*
- 2024** Julia Brzustowski, BSc internship, McMaster. *Flamegraphs for software maintenance.* (co-supervised with Vera Pantelic and Richard Paige)
 Cass Braun, B.Eng (NSERC USRA), McMaster. *Language Server Protocol for Justification diagrams.*
 Ahmed Elzaria, B.Eng McMaster. *Embedding Natural Language Processing into Mobile Applications to support privacy.*
 Natasha Lawford, B.Eng., McGill University. *Assurance Cases and Product Families.* (co-supervised with Alan Wassyng)
- 2023** Jonah Alle Monne (M.A.Sc, Université Grenoble Alpes). MITACS Globalink internship. *Exploring LLVM internal structure evolution over time.*
 Julia Brzustowski, BSc internship, McMaster. *Probes to extract information from legacy software.* (co-supervised with Vera Pantelic)
 Maël Charpentier, BSc internship (Université de Montréal). *Code completion for the P4 language.*
 Nirmal Chaudhari, B.Eng (NSERC USRA) McMaster. *Improving multi-language merge algorithms.*
 Ahmed Elzaria, B.Eng McMaster (Excellence in Research Award). *Investigating interactions among passes in the LLVM compiler toolchain.*
 Dennis Fong, B.Eng. McMaster. *Using SAT solving for package dependency management in SPACK.* (co-supervised with Camille Coti, ETS Montréal)
 Noel Chungath Gregory, B.Eng. McMaster. *Lightweight compiler engineering for the P4 language.*
 Aaron Loh. B.Eng. McMaster (Dean's Excellence list). *Analyzing DevOps CI pipelines at scale*
 Madhur Jain. (B.Sc., Indian Institute of Technology Bhilai). MITACS Globalink internship. *Improving multi-language support for Git-merge.*
 Nitish Kumar (B.Sc., Indian Institute of Technology Kharagpur). MITACS Globalink internship. *Identifying conflicting dependencies in SPACK.* (co-supervised with Camille Coti, ETS Montréal)
- 2022** Sathurshan Arulmohan, B.Eng. McMaster (Dean's Excellence list). *Using Natural Language Processing to extract conceptual models from user stories backlog.*
 Richard Li, B.Eng. internship (NSERC USRA). *Building a corpus of git merge conflicts.*
 Alexandre Niney B.Sc. internship. *Using AI to check game rules balance at scale* (co-supervised with Vladimir Reinharz)
 Floriane Paris, M.Eng internship. *Software visualization for version control system repositories.*
 Haotian Xe, M.Sc internship. *Graphical DSL for ArduinoML, a language to program the internet of things,* (co-supervised with Steffen Zschaler)

- Normand Lancelot, B.Sc. internship. *Measuring the Severity of the Signs of Eating Disorders Using Similarity-Based Models*. (co-supervised with Marie-Jean Meurs)
- 2021** Normand Lancelot, B.Sc. internship. *Extracting emotions from a twitter corpus*.
- Amélie Lachapelle-Dagenais, B.Sc. internship. *Adapting an application to the aging population*.
- 2020** Alyson Lecuyer, B.Tech. internship. *Showcasing students' result related to the aging population*.
- Avril de Goër de Herve, M.Sc. internship. *Impact analysis of compilation passes in LLVM*.
- Jérémy Fornarino, M.Eng. internship. *Collecting mental-health data from patients' phones*.
- Yan Conigliaro, M.Eng. internship. *Mining GitHub to build a corpus of conflicting merge scenarios*.
- Olivier Levasseur, B.Sc. internship. *Heuristics to improve git-merge for Java programs*.
- 2019** Chaima Frouni, B.Sc. internship. *A form-based approach to collect data from patients*.
- Gael Miton, Military engineering internship. *A simulator for underwater floating devices*.
- Mathieu Paillard, M.Eng. internship. *A DSL to support fast prototyping of composition operators*.
- Prune Pillone, M.Eng. internship. *Adapting software for the aging population*.
- Florian Juroszek, M.Eng. internship. *Static analysis of microservice architectures*.
- Alexis Segura, M.Eng. internship (Facebook Excellence Award). *Empirical analysis of git-merge conflicts*.
- Sébastien Michelland, M.Sc. internship. *Identifying conflicts in the LLVM toolchain*.
- 2018** Alexis Couvreur, M.Sc. internship. *Applying Smart contracts in an IoT context*.
- Florian Lehman, M.Eng. internship. *Software composition applied to Git*.
- Olivier Boulet, M.Eng. internship. *Securing sensor data collection using blockchain*.
- Florian Bourniquel, M.Eng. internship. *Visualizing interactions among code rewriters*.
- Johan Mortara, M.Eng. Internship. *Automated deployment of blockchain infrastructures*.
- 2016** Fabien Vicente, M.Eng. internship. *Containerizing a complex architecture: the Atlassian example*.
- Nicolas Lecourtois. M.Eng. internship. *Securing communications among containers*.

Research Assistants supervision

Start	End	Student	Position	Description
09/25	04/26	Andrew Bovbel	Research Assistant II	<i>Editor for the jPipe environment</i> .
05/25	08/25	Kalvin Khuu	Research Assistant III	<i>Development of composition operators for the jPipe compiler</i>
01/25	12/25	Nirmal Chaudhari	UQAM Research Scholarship	<i>Development of a mobile application using NLP to support mental health</i>
09/24	04/25	Cass Braun	Research Assistant II	<i>Maintenance of jPipe IDE</i>
09/24	04/25	Ahmed Elzaria	Research Assistant II	<i>Development of a mobile application using NLP to support mental health</i>
09/23	04/24	Nirmal Chaudhari	Research Assistant II	<i>Implementation of language support mechanisms for the jPipe language</i>
09/23	12/23	Noel Chungath Gregory	Research Assistant II	<i>Software maintenance of the p4-lsp open-source project</i> .

LIFETIME RESEARCH FUNDING

Note: When a grant was issued by a foreign funding agency and/or in another currency than CAD, an “approximative” translation to Canadian dollars is provided in addition to the original amount.

Ongoing Funding

Name(s) (indicate PI, underline your name)	Title/Purpose of Research	Years of Funding	Funding Source/Agency	Funding amount (by year)
J.M. Bruel (PI), <u>S. Mosser</u> (PI), M. Blay-Fornarino, P. Collet, F. Precioso,	FATES-MLOps: Incorporating FATES principles in continuous development of ML-integrated systems: a	2025 – 2028	NSERC Alliance International	\$90,000
		2024	Agence Nationale	€150,000

O. Teste, M. Pantel, R. Paige (co-apps)	MLOps perspective	– 2028	<i>de la Recherche</i> (ANR, France)	(~ \$250,000)
A. Wassyng (PI), M. Lawford (co-PI), <u>S. Mosser</u> (co-PI), R. Paige (co-app)	Model driven safety assurance for automotive over-the-air software updates	2023 – 2026	NSERC Alliance MITACS	\$265,000
Milena Head (PI) et al (17 co-applicants)	Aging, Mobility & the Digital Divide: Bridging Digital Divides for Older Adults Through Design	2024 – 2029	MIRA	\$200,000
F. Bordeleau (PI), J. Dingel, <u>S. Mosser</u> (co-PIs)	DevOps for Software Defined Network	2022 – 2025	NSERC-Mitacs Alliance Program	\$330,000
<u>S. Mosser</u> (PI)	Startup fund	N/A	Faculty of Engineering	\$200,000 (total)
<u>S. Mosser</u> (PI)	Software composition at large scale	2020 – 2026	NSERC Discovery Grant	\$29,000
		<i>1-year funded extension for co-chairing DG 1507</i>		

Funding Completed

Note: The “funding amount” column for the completed projects represents the total amount of funding for the project, for its whole duration (as, depending on funding agencies, installments might not be yearly-based).

Name(s) (indicate PI, underline your name)	Title/Purpose of Research	Years of Funding	Funding Source/Agency	Funding amount
M.-J. Meurs (PI), C. Bardon (PI), <u>S. Mosser</u> (co-app)	RÉSO-T : <i>Une approche innovante culturellement sensible pour intégrer le traitement automatique du langage naturel dans les outils de prévention du suicide</i>	2023 – 2024	FRQNT (Quebec)	\$50,000
S. Smith, D. Geiskovitch, R. Paige, S. Mosser (co-PIs)	Scalable Team-Based Learning - Structure Editor, GUI Editor and Teacher Dashboard	2023-2025 (20 months)	MITACS / STaBL Foundation	\$75,000 (total)
<u>S. Mosser</u> , R. Paige (co-PIs)	Centre of Excellence for Artificial Intelligence and Smart Mobility – RTA 5: Cloud Automation	2021 – 2026	Cubic Transportation Systems (CTS)	\$300,000
		<i>Project terminated in October 2023</i>		
Y. Farmer (PI), M.-E. Bouthillier, A. Duhoux, <u>S. Mosser</u> , M.-J. Meurs	<i>La perception populationnelle du risque sanitaire et l'acceptabilité sociale face au déconfinement. Informer les décideurs politiques à l'aide du forage de données sur Twitter</i>	<i>Project terminated in October 2023</i>	SSHRC	\$25,000
C. Messier (PI), M.-J. Meurs, J. Dupras, T. Handa, <u>S. Mosser</u> , A. Paquette, A. Smargiassi	<i>SylvCiT: un logiciel intelligent pour maximiser la résilience et les bienfaits des arbres municipaux face aux changements globaux</i>	2021 – 2022	FRQNT (QC)	\$140,000

S. Vial (PI), M-J Meurs, S. Gambs, S. Guay, <u>S. Mosser</u> (co-PIs)	<i>Mentallys, un service unique de cyber-santé mentale.</i>	2020 – 2022	FRQNT (Quebec)	\$100,000
M.-J. Meurs (PI), M. Benichou, G. Bondolfi, M. Bonenfant, S. Gambs, C. Malaterre, D. Martin, F. Millerand, <u>S. Mosser</u>	RELA: Respectful and Explainable AI to Support Struggling People and Mental Health Practitioners	2019 – 2022	NFRF Exploration	\$250,000
L. Gonnord, <u>S. Mosser</u> (co-PIs)	CharActerisation of Program Evolution with Static Analyses (CAPESA)	2019 – 2022	Inria (Équipe Associée)	30,000 EUR (~\$45,000)
<u>S. Mosser</u> (PI)	UQAM Faculty of Science Startup package	2019 – 2021	UQAM PAFARC	\$15,000 (unionized amount)
<u>S. Mosser</u> (co-PI), A.-M. Pinna-Déry (co-PI)	Software engineering for the aging population	2019 – 2021	FRQNT (QC), Ministère des affaires étrangères (MAE, FR)	\$18,300 + 10,000 EUR (~\$33,000)
L. Lizzi (PI) <i>et al</i> (8 co-applicants)	Internet of Things Wireless Infrastructures (I-Win).	2018 – 2019	UCA Initiative of Excellence	36,000 EUR (~\$54,000)
F. Verdier (PI) <i>et al</i> (6 co-applicants)	Smart IoT for mobility (Phase I).	2018 – 2019	UCA Initiative of Excellence	25,000 EUR (~\$37,500)
<u>S. Mosser</u>	Formalising Scalable Composition Operators (FIASCO)	2018	CNRS Research Accelerator	5,000 EUR (~\$7,500)
M. Blay-Fornarino (co-PI), <u>S. Mosser</u> (co-PI), G. Nolet	Software Composition for the MERMAID	2017 – 2020	Provence - Alpes Côte d'Azur regional research fund	100,000 EUR (~\$150,000)
P. Collet (co-PI), <u>S. Mosser</u> (co-PI).	Variability in cyber-Physical Systems	2016 – 2020	Industrial contract (VISTEON)	100,000 EUR (~\$150,000)
B. Benni, <u>S. Mosser</u> (PI)	Modelling Software Composition	2016 – 2019	UCA school of graduate studies	100,000 EUR (~\$150,000)
<u>S. Mosser</u> (PI)	Modelling for scaling (M4S)	2016	CNRS early career accelerator	10,000 EUR (~\$15,000)
C. Cecchinel, P. Collet (co-PI), <u>S. Mosser</u> (co-PI)	DEPOSIT at scale	2017	European Institute of Innovation and Technology (EIT Digital), industrial transfer program.	35,000 EUR (~\$52,500)
C. Cecchinel, P. Collet (co-PI), <u>S. Mosser</u> (co-PI)	Tailored composition for large-scale sensing networks	2014 – 2017	UCA school of graduate studies	100,000 EUR (~\$150,000)
I. Logre, <u>S. Mosser</u> (PI)	Model-based sensor data visualizations	2013 – 2017	UCA school of graduate studies	100,000 EUR (~\$150,000)

M. Blay-Fornarino (co-PI), <u>S. Mosser</u> (co-PI)	Domain-specific languages & Software Product Line for Cloud-computing (IDOL)	2012 – 2014	European Union international cooperation research fund (EGIDE), Aurora program	20,000 EUR (~\$30,000)
<u>S. Mosser</u> (PI)	Modelling for Cloud-computing	2012 – 2014	Amazon research sponsorship	25,000 EUR (~\$37,500)
M. Blay-Fornarino (PI)	YourCast, an <i>a-la-carte</i> information broadcasting system	2012 – 2014	<i>Agence nationale de la recherche</i> (ANR), Technological Transfer program	250,000 EUR (~\$375,000)
E. di Nitto (PI) <i>et al</i> (12 universities)	Model-driven approach for design and execution of applications on multiple clouds (MODAClouds)	2011 – 2015	European Union Research Fund, Framework Program 7 (EU-FP7)	8,700,000 EUR (~\$13,000,000)
Keith Jeffery (PI) <i>et al</i> (17 universities)	A model-based cross cloud development and deployment platform (PaaSage).	2011 – 2016	European Union Research Fund, Framework Program 7 (EU-FP7)	9,700,000 EUR (~\$14,500,000)
G. Horn (PI) <i>et al</i> (13 universities)	Reuse and Migration of legacy applications to interoperable cloud services (REMICS)	2011 – 2016	European Union Research Fund, Framework Program 7 (EU-FP7)	4,500,000 EUR (~\$6,750,000)
A. Solberg (PI) <i>et al</i> (12 universities)	Environmental services infrastructure with ontologies (ENVISION)	2010 – 2013	European Union Research Fund, Framework Program 7 (EU-FP7).	\$4,500,000 (~\$6,750,000)

Funding Applied for

Name(s) (indicate PI, underline your name)	Title/Purpose of Research	Years of Funding	Funding Source/Agency	Funding amount (by year)
<u>S. Mosser</u> (PI), 7 others co-PIs (4 universities)	Ensuring safety while developing mental-health-related applications	5	NSERC Alliance Society	\$300,000
<i>Status: Consortium finalizing the proposal. To be submitted Fall 2025.</i>				
R. Paige (PI), D. Down (co-PI), S. Mosser (co-PI), V. Pantelic (co-PI)	Automated Software and Performance Engineering for Integrated Transportation Systems	2024 – 2028	NSERC Alliance	\$785,000
<i>Status: Aborted. Partner terminated contract with McMaster for financial reasons.</i>				

LIFETIME PUBLICATIONS

Institution	Year	Journals		Conferences		Workshops	
		International	National	International	National	International	National

McMaster	2025	2	1	3			
	2024	1	3	3			
	2023	2	1	5			
	2022	2	3				
UQAM	2021	1	3				
	2020	3	4				
	2019	3	2				
UCA	2018	2	5	5			
	2017	In charge of the redesign of the software engineering entire curriculum at Polytech Sophia					
	2016		2				
	2015		1	1			
	2014		2	3			
	2013	1	1	4			
SINTEF	2012		4	1	5		
Inria	2011		5	2	1		
U. Nice (Ph.D.)	2010	1	1	1			
	2009		1	1	1		
	2008		2	1	1		
	2007	1	1	1	1		
Total:		16	1	39	7	24	2

Underlined names are students under my direct supervision for the work done in the publication.

Peer Reviewed

Journal Articles

- [J1] N. Selmi, J.-M. Bruel, **S. Mosser**, M. Crespo, A. Kerbrat. Engineering Decisions in MBSE: Insights for a Decision Capture Framework Development. *INSIGHT journal, International Council on System Engineering (INCOSE)*. 2025
- [J2] R. Jongeling, **S. Mosser**. Preface to the special issue on the 21st European Conference on Modelling Foundations and Applications (ECMFA 2025). *Journal of Object Technology (JOT)*, Vol. 24, n°2. May 2025.
- [J3] G. Mussbacher, B. Combemale, J. Kienzle, L. Burgueño, A. García-Domínguez, J.-M. Jézéquel, G. Jouneaux, D. E. Khelladi, **S. Mosser**, C. Pulgar, H. A. Sahraoui, M. Schiedermeier, T. vd Storm: Polyglot Software Development: Wait, What? *IEEE Software*. 41(4): 124-133 (2024)
- [J4] M. Levy, E. C. Groen, K. Taveter, D. Amyot, E. Yu, L. Liu, I. Richardson, M. Spichkova, A. Jussli, and **S. Mosser**. Sustaining Human Health: A Requirements Engineering Perspective. *Journal of Systems and Software (JSS)*, 204, pages 111792, 2023,
- [J5] W. K.G. Assunção, J. Krüger, **S. Mosser**, and S. Selaoui. How do microservices evolve? An empirical analysis of changes in open-source microservice repositories. *Journal of Systems and Software (JSS)*, 204, pages 111788, 2023.
- [J6] I. Trabelsi, M. Abdellatif, A. Abubaker, N. Moha, **S. Mosser**, S. Ebrahimi-Kahou, and Y.-G. Guéhéneuc. From legacy to microservices: A type-based approach for microservices identification using machine learning and semantic analysis. *Journal of Software: Evolution and Process (JSEP)*, 2022.
- [J7] **S. Mosser**, V. Reihnarz, and C. Pulgar. Modelling Agile Backlogs as Composable Artefacts to support Developers and Product Owners. *Journal of Object Technology (JOT)*. 2022
- [J8] B. Combemale, J. Kienzle, G. Mussbacher, H. Ali, D. Amyot, M. Bagherzadeh, E. Batot, N. Bencomo, B. Benni, J.-M. Bruel, J. Cabot, B. H. C. Cheng, P. Collet, G. Engels, R. Heinrich, J.-M. Jézéquel, A. Koziolk, **S. Mosser**, R. H. Reussner, H. A. Sahraoui, R. Saini, J. Sallou, S. Stinckwich, E. Syriani, and M. Wimmer. A Hitchhiker's Guide to Model-Driven Engineering for Data-Centric Systems. *IEEE Software*. 2021
- [J9] S. Bonnioux, D. Cazau, **S. Mosser**, M. Blay-Fornarino, Y. Hello, and G. Nolet. MeLa: A Programming Language

- for a New Multidisciplinary Oceanographic Float. *MDPI Sensors*, 2020.
- [J10] **B. Benni**, **S. Mosser**, M. Acher, and M. Paillart. Characterizing Black-box Composition Operators via Generated Tailored Benchmarks. *Journal of Object Technology (JOT): special issue ECMFA '20*, June 2020.
- [J11] G. Mussbacher, B. Combemale, J. Kienzle, S. Abrahão, H. Ali, N. Bencomo, M. Búr, L. Burgueño, G. Engels, P. Jeanjean, J.-M. Jézéquel, T. Kühn, **S. Mosser**, H. Sahraoui, E. Syriani, D. Varró, and M. Weyssow. Opportunities in Intelligent Modeling Assistance. *Software and Systems Modeling*, 2020.
- [J12] **C. Cecchinell**, F. Fouquet, **S. Mosser**, and P. Collet. Leveraging live machine learning and deep sleep to support a self-adaptive efficient configuration of battery powered sensors. *Future Generation Computer Systems (FGS)*, Mar. 2019.
- [J13] **B. Benni**, **S. Mosser**, N. Moha, and M. Riveill. A Delta-oriented Approach to Support the Safe Reuse of Black-box Code Rewriters. *Journal of Software: Evolution and Process (JSEP), ICSR special issue*, July 2019.
- [J14] L. Burgeno, F. Ciccozzi, M. Famelis, G. Kappel, L. Lambers, **S. Mosser**, R. Paige, A. Pierantonio, A. Rensink, R. Salay, G. Taentzer, A. Vallecillo, and M. Wimmer. Contents for a Model-Based Software Engineering Body of Knowledge. *Journal of Software and Systems Modeling*, June 2019.
- [J15] **S. Lazreg**, P. Collet, and **S. Mosser**. Functional Feasibility Analysis of Variability-Intensive Dataflow-oriented Applications over Highly configurable Platforms. *ACM SIGAPP Applied Computing Review*, Sept. 2018.
- [J16] B. Combemale, J. Kienzle, G. Mussbacher, O. Barais, E. Bousse, W. Cazzola, P. Collet, T. Degueule, R. Heinrich, J.-M. Jézéquel, M. Leduc, T. Mayerhofer, **S. Mosser**, M. Schöttle, M. Strittmatter, and A. Wortmann. Concern-Oriented Language Development (COLD): Fostering Reuse in Language Engineering. *Computer Languages, Systems and Structures*, 2018.
- [J17] **S. Mosser** and M. Blay-Fornarino. ADORE, a Logical Meta-model Supporting Business Process Evolution. *Science of Computer Programming*, 78(8):1035 – 1054, 2013.
- [J18] **S. Mosser**, M. Blay-Fornarino, and R. France. Workflow Design using Fragment Composition (Crisis Management System Design through ADORE). *Transactions on Aspect-Oriented Software Development (TAOSD)*, Special issue on Aspect Oriented Modeling:1–34, 2010.
- [J19] M. Blay-Fornarino, V. Hourdin, C. Joffroy, S. Lavirotte, **S. Mosser**, A.-M. Pinna Déry, P. Renevier, M. Riveill, and J.-Y. Tigli. Architecture pour l’adaptation de Systèmes d’Information Interactifs Orientés Services. *Revue des Sciences et Technologies de l’Information - Série L’Objet : logiciel, bases de données, réseaux*, pages 93–118, 2007.

Other (Proceedings of International Conferences)

- [C1] B. Combemale, J. Kienzle, G. Mussbacher, P. Archambault, J.-M. Bruel, L. Burgueño, B. H.C. Cheng, L. Cleophas, G. Engels, D. Foures, S. Klikovits, V. Kulkarni, J. Michael, **S. Mosser**, H. Sahraoui, E. Syriani, A. Wortmann. On the Challenges of Integrating Digital Twins. *2nd International Conference on Engineering Digital Twins (EDTconf 2025)*, co-located with *ACM/IEEE International Conference on Model Driven Engineering Languages and Systems (MODELS 25)*
- [C2] **S. Mosser**, **N. Chaudhari**, **C. Braun**, **K. Sun**. Creating and Operationalizing Justification Models Using jPipe (tutorial). *ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion*.
- [C3] J.-M. Bruel and **S. Mosser**. Modern Teaching of Requirements Engineering and Business Analysis (tutorial). In *International Requirements Engineering Conference*, RE, Sep. 2024.
- [C4] D. Maupomé, Y. Ferstler, **S. Mosser**, M.-J. Meurs. Automatically Finding Evidence and Predicting Answers in Mental Health Self-Report Questionnaires. *CLEF (Working Notes) 2024*: 841-850
- [C5] D. Maupomé, T. Soulas, F. Rancourt, G. Cantin-Savoie, G. Winterstein, **S. Mosser**, and M.-J. Meurs. Lightweight methods for early risk detection. In *Proceedings of the Working Notes of CLEF 2023 - Conference and Labs of the Evaluation Forum, Thessaloniki, September 18th - to - 21th, 2023 (CEUR Workshop Proceedings)*, 2023.
- [C6] S.H. Hosseini Saravani, **L. Normand**, D. Maupomé, F. Rancourt, T. Soulas, S. Besharati, A. Normand, **S. Mosser**, and M.-J. Meurs. Measuring the Severity of the Signs of Eating Disorders Using Similarity-Based Models. In *Proceedings of the Working Notes of CLEF 2022 - Conference and Labs of the Evaluation Forum, Bologna, Italy, September 5th - to - 8th, 2022 (CEUR Workshop Proceedings)*, *CEUR-WS.org*, 3180, pages 936-946, 2022.
- [C7] J. Kienzle, B. Combemale, G. Mussbacher, O. Alam, F. Bordeleau, L. Burgueño, G. Engels, Jessie J., J.-M. Jézéquel, B. Kemme, **S. Mosser**, H. A. Sahraoui, M. Schiedermeier, and E. Syriani. Global Decision Making

- Over Deep Variability in Feedback-Driven Software Development. In *37th IEEE/ACM International Conference on Automated Software Engineering, ASE 2022, Rochester, MI, USA, October 10-14, 2022 ACM*, pages 178:1-178:6, 2022.
- [C8] J. Krüger, W. K. G. Assunção, I. Ayala, and **S. Mosser**. International Workshop on Variability Management for Modern Technologies (VM4ModernTech 2022). In *SPLC '22: 26th ACM International Systems and Software Product Line Conference, Graz, Austria, September 12 - 16, 2022, Volume A ACM*, pages 266, 2022.
- [C9] A. Lapointe-Boisvert, **S. Mosser**, and S. Trudel. Towards Modelling Acceptance Tests as a Support for Software Measurement. In *13th System Analysis and Modelling Conference - ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion, MODELS 2021 Companion, Fukuoka, Japan, October 10-15, 2021 IEEE*, pages 827-832, 2021.
- [C10] A. Lachapelle-Dagenais, **S. Mosser**, A.-M. Pinna-Dery, and M. Blay-Fornarino. Requirements Engineering for the Ageing Population: a Teaching Perspective. In *29th IEEE International Requirements Engineering Conference Workshops, RE 2021 Workshops, Notre Dame, IN, USA, September 20-24, 2021 IEEE*, pages 248-257, 2021.
- [C11] **S. Mosser** and J.-M. Bruel. Requirements Engineering in the DevOps Era (tutorial). In *International Requirements Engineering Conference, RE*, Sep. 2021.
- [C12] **S. Mosser**, J.-P. Caissy, F. Juroszek, F. Vouters, and N. Moha. Charting Microservices to Support Services' Developers: the Anaximander Approach. In *International Conference on Service-Oriented Computing (ICSOC)*, short paper, Dec. 2020.
- [C13] G. Mussbacher, B. Combemale, S. Abrahão, N. Bencomo, L. Burgueño, G. Engels, J. Kienzle, T. Kühn, **S. Mosser**, H. Sahraoui, and M. Weyssow. Towards an Assessment Grid for Intelligent Modeling Assistance. In *MDE Intelligence 2020 - 2nd Workshop on Artificial Intelligence and Model-driven Engineering*, Oct. 2020.
- [C14] D. Maupomé, M. D. Armstrong, R. M. Belbahar, J. Alezot, R. Balassanio, M. Queudot, **S. Mosser**, and M.-J. Meurs. Early mental health risk assessment through writing styles, topics and neural models. In *Working Notes of CLEF 2020 - Conference and Labs of the Evaluation Forum*, 2020.
- [C15] B. Benni, **S. Mosser**, J.-P. Caissy, and Y.-G. Guéhéneuc. Can Microservice-Based Online-Retailers be Used as an SPL? In *International System and Software Product Line Conference (SPLC)*, Dec. 2020.
- [C16] S. Lazreg, M. Cordy, P. Collet, P. Heymans, and **S. Mosser**. Multifaceted Automated Analyses for Variability-Intensive Embedded Systems. In *41st ACM/IEEE International Conference on Software Engineering*, ICSE, May 2019.
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- [C19] B. Benni, **S. Mosser**, N. Moha, and M. Riveill. A Delta-oriented Approach to Support the Safe Reuse of Black-box Code Rewriters. In *17th International Conference on Software Reuse (ICSR'18)*, Madrid, France, May 2018.
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- [C22] F. Fouquet, T. Hartmann, **S. Mosser**, and M. Cordy. Enabling lock-free concurrent workers over temporal graphs composed of multiple time-series. In *Symposium on Applied Computing*, volume 8, Pau, France, Apr. 2018.
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- [C27] I. Logre, S. Mosser, P. Collet, and M. Riveill. Sensor Data Visualisation: A Composition-Based Approach to Support Domain Variability. In *European Conference on Modelling Foundations and Applications (ECMFA 2014)*, volume 8569, pages 101–116, York, United Kingdom, July 2014. Springer.
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- [C38] S. Mosser, A. Bergel, and M. Blay-Fornarino. Visualizing and Assessing a Compositional Approach of Business Process Design. In *Software Composition 2010*, page Springer's Lecture Notes in Computer Science, Malaga, Spain, June 2010. ACM SIGPLAN and SIGSOFT.
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- [C40] S. Mosser, F. Chauvel, M. Blay-Fornarino, and M. Riveill. Web Service Composition: Mashups Driven Orchestration Definition. In *International Conference on Intelligent Agents, Web Technologies and Internet Commerce (IAWITC'08)*, pages 1–6, Vienna, Austria, Dec. 2008. IEEE Computer Society.
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Other (Proceedings of International Workshops)

- [W1] N. Selmi, J.-M. Bruel, S. Mosser and M. Crespo. Towards Model-Based Decision-Telling: Design Evolution Through Decision Nodes. In *18th Workshop on Models and Evolution (ME, co-located with MODELS)*. Oct 2025

- [W2] J.-M. Bruel, T. Gouaichault, O. Teste, M. Blay-Fornarino, N. Lacroix, F. Precioso, **S. Mosser**. Incorporating FATES Principles in Continuous Development of ML-Integrated Systems: Importance of Requirements. In *1st International Workshop on Requirements Engineering for Accountable and Conscious Human-centered AI (REACH'AI)*, co-located with *IEEE RE 2025*. Valencia, Spain. September 2025.
- [W3] T. Camargo da Silva, L. Lambers, **S. Mosser** and K. Revoredo. Provider-Agnostic Knowledge Graph Extraction from User Stories using Large Language Models. In *First Workshop on Large Language Models For Generative Software Engineering (LLM4SE 2025)*, co-located with *STAF 2025*, Koblenz, Germany, June 2025.
- [W4] M. Blay-Fornarino, J.-M. Bruel, **S. Mosser**, F. Precioso. Prise en compte des propriétés FATES en MLOps: perspectives et ambitions. In *4th edition of EXPLAIN'AI*, CNRS. January 2025.
- [W5] A. Lachance, **S. Mosser**. Building deduplicated model repositories to assess domain-specific languages evolution. In *17th Workshop on Models and Evolution (ME)*, co-located with *MODELS*. September 2024.
- [W6] T. Chiang, R. Paige, A. Wassyng, **S. Mosser**. Mapping Requirements to Features to Create Traceability in Product Line Models. In *6th International Workshop on Modeling Language Engineering (MLE)*, co-located with *MODELS*. September 2024.
- [W7] T. Chiang, R. Paige, A. Wassyng, **S. Mosser**. A Tool For Feature-Requirement Traceability Using Requirement Canvas and Encapsulation. In *5th Modeling in Automotive System and Software Engineering Workshop (MASE)*, co-located with *MODELS*. September 2024.
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- [W9] S. Arulmohan, M.-J. Meurs, **S. Mosser**. Extracting Domain Models from Textual Requirements in the Era of Large Language Models. In *5th Workshop on Artificial Intelligence and Model-driven Engineering (MDEIntelligence)*, col-located with *MODELS*. October 2023
- [W10] A. Bucchiarone, A. Vazquez-Ingelmo, G. Schiavo, S. Barandoni, A. Garcia-Holgado, F.J. Garcia-Penalvo, **S. Mosser**, A. Pierantonio, S. Zschaler, W. Barnett. Towards Personalized Learning Paths to Empower Competency Development in Model Driven Engineering through the ENCORE platform. In *26th International Conference on Model-Driven Engineering Languages and Systems (MODELS)*, *Educator Symposium*. 2023.
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- [W21] P. Collet, **S. Mosser**, S. Urli, M. Blay-Fornarino, and P. Lahire. Experiences in Teaching Variability

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- [W30] D. Ardagna, E. Di Nitto, G. Casale, D. Petcu, P. Mohagheghi, **S. Mosser**, P. Matthews, A. Gericke, C. Ballagny, F. D'Andria, C.-S. Nechifor, and C. Sheridan. MODAClouds: A Model-driven Approach for the Design and Execution of Applications on Multiple Clouds. In *Proceedings of the 4th International Workshop on Modeling in Software Engineering, MiSE '12*, pages 50–56, Piscataway, NJ, USA, 2012. IEEE Press.
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- [W33] **S. Mosser**. Are Functional Languages a good way to represent productive meta-models? In *4th European Lisp Workshop (ELW'07)*, pages 1–6, Berlin, Germany, France, July 2007.

Other (Proceedings of National Conferences and Workshops)

- [N1] F. Chauvel, **S. Mosser**, and A. Solberg. Reconsidering QoS Analysis in Dynamic and Open Systems. In *1ère conférence en ingénierie du logiciel (CIEL'12), short paper*, Rennes, June 2012.
- [N2] A. Feugas, **S. Mosser**, A.-F. Le Meur, and L. Duchien. Déterminer l'impact d'une évolution dans les processus métiers. In *Journées sur l'Ingénierie Dirigée par les Modèles (IDM'11)*, pages 71–76, Lille, France, June 2011.
- [N3] C. Brel and **S. Mosser**. Vers une approche flot de données pour supporter la composition d'interfaces homme-machine. In *Journées sur l'Ingénierie Dirigée par les Modèles (IDM'11)*, pages 1–6, Lille, France, June 2011. CNRS.
- [N4] **S. Mosser** and M. Blay-Fornarino. Taming Orchestration Design Complexity through the ADORE Framework. In *Journées 2010 du GDR GPL, CNRS*, Pau, France, Mar. 2010.
- [N5] **S. Mosser** and M. Blay-Fornarino. Réflexions autour de la construction dirigée par les modèles d'un atelier de composition d'orchestrations. In *15ème conférence francophone sur les Langages et Modèles à Objets (LMO'09)*, pages 1–16, Nancy, France, Mar. 2009. Cépadués.

- [N6] **S. Mosser**, M. Blay-Fornarino, and M. Riveill. Un modèle d'évolution multi-vues des Architectures Orientées Services. In *Actes de l'Atelier Doctorant LMO'08(DOC LMO'08), workshop*, , page 6, Montréal, Mar. 2008. Université de Montréal -.
- [N7] **S. Mosser**, M. Blay-Fornarino, P. Collet, and P. Lahire. Vers l'intégration dynamique de contrats dans des architectures orientées services : une expérience applicative du modèle au code. In *2ème Conférence sur les Architectures Logicielles (CAL'08)*, pages 1–15, Montréal, Canada, Mar. 2008.
- [N8] **S. Mosser**, M. Blay-Fornarino, and M. Riveill. Orchestrations de Services Web : Vers une évolution par composition. In *Atelier RIMEL (Rétro-Ingénierie, Maintenance et Evolution des Logiciels)*, , page 6, Toulouse, France, Mar. 2007. Dalila Tamzalit, Salah Sadou.
- [N9] C. Joffroy, **S. Mosser**, M. Blay-Fornarino, and C. Nemo. Des Orchestrations de Services Web aux Aspects. In U. d. T. EMN, INRIA, editor, *3ème Journée Francophone sur le Développement de Logiciels Par Aspects (JFLDPA'2007)*, pages 1–13, Toulouse, France, Mar. 2007.

Non-Peer Reviewed

Community Engagement and Knowledge Exchange

- "Qualité Logicielle: Retour sur le bug de l'an 2000, 25 ans plus tard". Interview pour le documentaire Le Bogue 2000, TV-Québec. October 2024.
- "Des fonctionnaires fédéraux trans épuisés de « se battre » pour changer leur identité". Interview for Radio-Canada (on the complexity of evolving legacy software), March 2024.
- "Banff, le lieu de rendez-vous des ingénieurs exigeants". Interview for Radio-Canada (on the importance of requirements engineering for software development), August 2018.
- **S. Mosser**. *La Thèse ...* (seminar for new Ph.D. Students at Inria Lille – Nord Europe). 2011.

Submitted for Publication

- N/A

PRESENTATIONS AT MEETINGS

Keynotes

- [K1] **S. Mosser**. Justifying Machine Learning Pipelines: Natural Language Processing and Agility. Café FSL. Institut de Recherche en Informatique de Toulouse (IRIT, CNRS). February 2025.
- [K2] **S. Mosser**. Software Engineering and Natural Language Processing: What Can Go Wrong? CRIM Seminars, Montreal, November 2024.
- [K3] A. Lachance and **S. Mosser**. Developing a modular language server to support P4 developers. P4 developers days meeting, September 2023.
- [K4] **S. Mosser**. From Software Composition at Scale to Scaling software composition: 50 shades of scalability. Consortium for Software Engineering Research (CSER) 2022 Spring Meeting. May 2022.
- [K5] B. Benni and **S. Mosser**. Applying Software Composition to the Docker Ecosystem. Amadeus Global Tech Forum. **Keynote**. Oct. 2018
- [K6] **S. Mosser**. Renforcer l'engagement étudiant en projet. Journées sur la pédagogie active, Université Bretagne-Loire. **Keynote**. July 2017.
- [K7] **S. Mosser**. Projets, Agilité & École d'Ingénieur. Journées sur l'Innovation Pédagogique, Université du Maine. **Keynote**. Mar. 2017.

Invited

- [P1] N. Chaudhari, **S. Mosser**. Modelling justification diagrams using jPipe. The MDE Network, June. 2024
- [P2] A. Lachance, **S. Mosser**. From Zero to VS Code: A Framework Approach to Language Support. The MDE Network, October. 2023
- [P3] **S. Mosser**. Teaching Modelling, Modelling Teaching. The MDE Network, Mar. 2023.
- [P4] **S. Mosser**. Dockerizing your Teaching: Do's and Don'ts. The MDE Network, Mar. 2022
- [P5] **S. Mosser**. Software Composition in a Cyber-Physical World. Canada-Norway collaboration,

Østfold College, Mar 2022.

- [P6] **S. Mosser.** *Génie Logiciel pour la Population Vieillissante*. La France à l'UQAM, Feb. 2022.
- [P7] **S. Mosser.** Software Composition for the IoT & Cloud. Canada Border Services Agency (PD&DD, BTID), Dec. 2021.
- [P8] **S. Mosser.** Justification Diagrams in a DevOps Context. Model-driven Engineering & Requirements Engineering working groups, CNRS. Dec. 2021.
- [P9] **S. Mosser.** Building a CI/CD pipeline (demo). *Association Générale des Étudiantes et Étudiants en Informatique de l'UQAM* (Invited seminar). Dec. 2021.
- [P10] **S. Mosser.** *User stories & Acceptance Testing*. Canada Border Services Agency (PD&DD, BTID), Nov. 2021.
- [P11] **S. Mosser.** Building Software for the Ageing Population: A Software Engineering Point of View. Smart Mobility for the ageing Population (sMAP) research seminar, Canada. Oct. 2021.
- [P12] **S. Mosser.** Docker in a CI/CD context. Canada Border Services Agency (PD&DD, BTID), Oct. 2021.
- [P13] **S. Mosser.** Anaximander, a lightweight approach to support software exploration. Working group on software adaptation (YODA), *Centre National de la Recherche Scientifique* (CNRS). Feb. 2021.
- [P14] **S. Mosser.** Using a project-based approach to support Software Engineering teaching. LATECE seminar, UQAM, Montréal, Canada. Feb. 2020.
- [P15] **S. Mosser.** How can models help data scientists? Lessons learned from an undercover agent. 2nd Winter Modelling Meeting. San Vigilo de Marrebe, Italy. Feb. 2020.
- [P16] **S. Mosser.** Software Composition in a Cyber-Physical World. Ptidej Research Seminar, Concordia University, Montréal, Canada. Dec. 2019.
- [P17] **S. Mosser.** Les aspects génie logiciel pour les Systèmes Cyber-Physique. In *Journées IIoT du GDR MACS*, CNRS, France, July 2018.
- [P18] V. Aranega, A. Etien, and **S. Mosser.** Using Feature Model to build Model Transformation Chains. In *Journées 2013 du GDR GPL*, CNRS, France, Mar. 2013.
- [P19] **S. Mosser**, G. Mussbacher, M. Blay-Fornarino, and D. Amyot. Une approche orientée aspect allant du modèle d'exigences au modèle de conception. In *Journées du GDR GPL*, pages 37–38, Lille, France, June 2011.

PATENTS, INVENTIONS AND COPYRIGHTS

- 2014 “*Assets logiciels utilisés pour réaliser un Système de Diffusion d'Information - YourCast*”. Mireille Blay-Fornarino, Simon Urli, **Sébastien Mosser** and Daniel Romero. *Agence de la Protection des Programmes* (APP) IDDN.FR.001.320001.000.S.C.2014.000.31235, France.
- 2010 “*Diffusion d'informations par composition - JSEDUITE*”. **Sébastien Mosser**, Mireille Blay-Fornarino, Michel Riveill and David Emsellem. *Agence de la Protection des Programmes* (APP) IDDN.FR.001.120009.000.S.P.2011.000.00000, France.

SOFTWARE AND DATASETS

- 2025 P. Collet, **S. Mosser**, D. Wang. *Architecture Decision Records Dataset*. <https://ace-design.github.io/island/>
- 2024 Island. *Serious game to teach Software Engineering*. <https://ace-design.github.io/island/>
- 2023 S. Arulmohan, **S. Mosser** and M.-J. Meurs. *Qualified user stories (ground truth, Visual Narrator, GPT-3.5, CRF)*. Version 1.0 (11/07/2023). <http://doi.org/10.5281/zenodo.8136975>
- 2023 *jPipe. A software language to justify CICD pipelines*. <https://github.com/ace-design/jpipe>
- 2023. *p4-lsp. A language server to support P4 developers*. <https://github.com/ace-design/p4-lsp>

ADMINISTRATIVE RESPONSIBILITIES

Department

06/25 – 05/28	Tenure and Promotion Committee	(re-elected, 3-year)
07/24 – ...	Associate Chair, Undergrad (Internal).	(appointed, 3-year)
07/24 – ...	Equity, Diversity, Inclusion Committee	(chair)
07/24 – ...	Software Engineering Curriculum Committee	(member)

07/23 – 06/25	Tenure and Promotion Committee	(elected, interim, 2-year)
05/24 – 06/24	Department manager hiring committee	(member)
11/23 – 03/24	CAS Chair search committee	(elected)
07/23 – 06/24	Software Engineering Curriculum Committee	(chair)
01/22 – 06/24	Undergraduate advisor for Software Engineering	(appointed)
05/23 – 08/23	Hiring committee, teaching-track & CLA	(member)
01/22 – 06/23	Software Engineering Curriculum Committee	(member)

Faculty

12/24 – ...	Associate Director, McSCert	(appointed)
09/24 – 12/24	McSCert Director Search Committee	(appointed)
11/22 – 01/23	Ad hoc Selection Committee, Associate Dean – Academic.	(appointed)

University

01/2024 – ...	Expert Panel on Artificial Intelligence Research.	(member)
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ADMINISTRATIVE RESPONSIBILITIES OUTSIDE OF MCMASTER (until 2022)

Department

2020 – 2021	Co-chair of the M.Eng. in Software Engineering (UQAM). <i>Elected.</i>
2019 – 2021	Deputy chair of the B.Sc. in Comp. Science and Soft. Engineering (UQAM). <i>Elected.</i>
2014 – 2018	Computer Science Department Executive Board (UCA). <i>Elected.</i>
2013 – 2018	Director of the M.Sc. in Software Architecture (UCA). <i>Appointed.</i>
2012 – 2018	Coordinator of project-based teaching for software engineering (UCA). <i>Appointed.</i>

Faculty

2018	Executive board of the Computer Science Research Center (I3S, UCA/CNRS). <i>Appointed.</i>
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University

2020 – 2021	<i>Comité Apprentissage Recherche</i> (CAR, advising on digital strategy & IT). UQAM
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OTHER RESPONSIBILITIES

SE@MTL	Together with J. Kienzle (McGill), F. Bordeleau (ÉTS) and H. Sahraoui (UdeM), we founded in 2019 the <i>Software Engineering at Montreal</i> community to animate the local research ecosystem by organizing monthly seminars that bring together ~30 participants
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